

STRENGTH OF MATERIALS

Complex Stress

Stress on Inclined Plane/ Stress Transformation/

CONTENTS

1.1 Introduction

1.2 Stresses on Oblique Plane

1.3 Principal Stresses and Maximum Shear Stress

1.4 Mohr's Circle of Stress

OBJECTIVES

At the end of this lesson students will be able to:

- Evaluate stresses on any plane through stress-transformation equations.
- Evaluate principal stresses and locate principal planes for plane stress state
- Find maximum shear stresses at a point on a body for plane stress state
- Construct Mohr's circle for stress and identify the points of principal stress and maximum shear stresses

KEY TERMS

- Plane stress
- Complex stress
- Stress transformation
- Stress transformation equations
- Principal stress
- Principal plane

1.1 Introduction

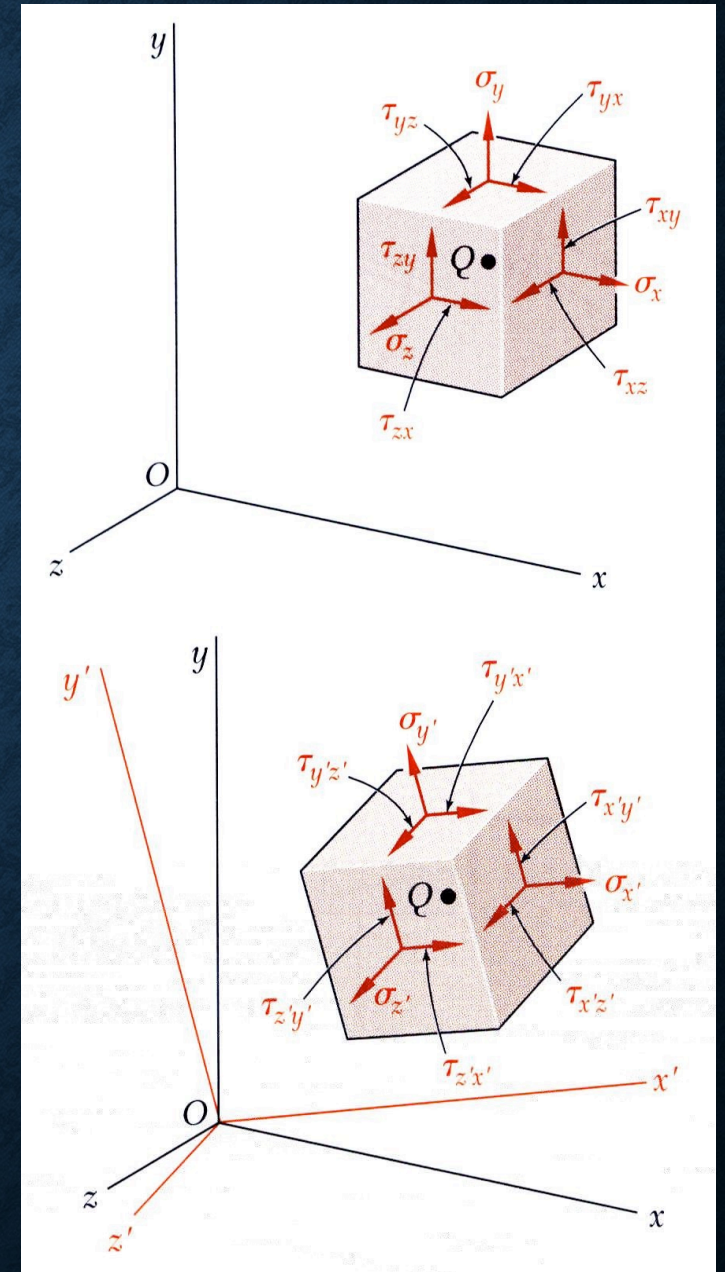
General state of stress:

➤ The most general state of stress *at a given point Q* may be represented by 6 components

- Three Normal stresses ($\sigma_x, \sigma_y, \sigma_z$) and
- Three Shearing stresses ($\tau_{xy}, \tau_{yz}, \tau_{zx}$)

(Note that $\tau_{xy} = \tau_{yx}, \tau_{yz} = \tau_{zy}, \tau_{zx} = \tau_{xz}$)

➤ Same state of stress is represented by a different set of components if axes are rotated



PLANE STRESS

- This section is concerned with a special two-dimensional state of stress called **plane stress**.
- It is important for two reasons:
 - (1) It arises in real components (particularly in thin components loaded in certain ways), and
 - (2) It is a two dimensional state of stress, and thus serves as an excellent introduction to more complicated three dimensional stress states.

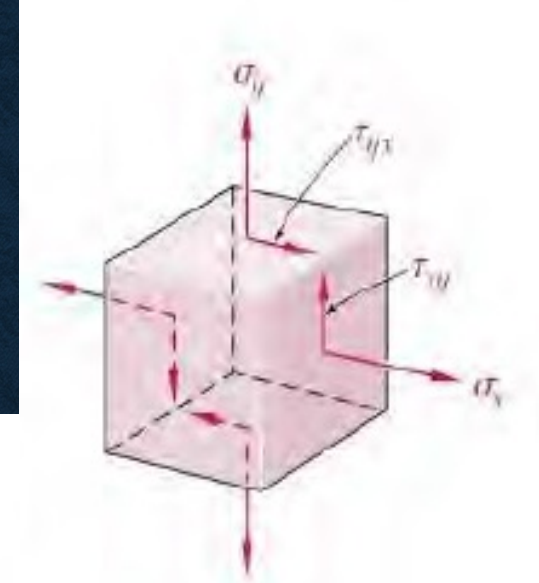
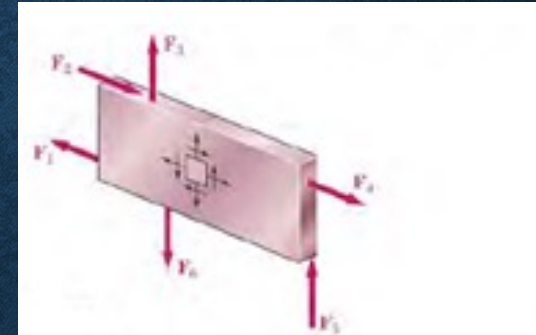
Plane Stress:

➤ A situation in which two of the faces of the cubic element are free of any stress

- For the previous general state of stress case,
the state of plane stress is defined by σ_x , σ_y , τ_{xy} and

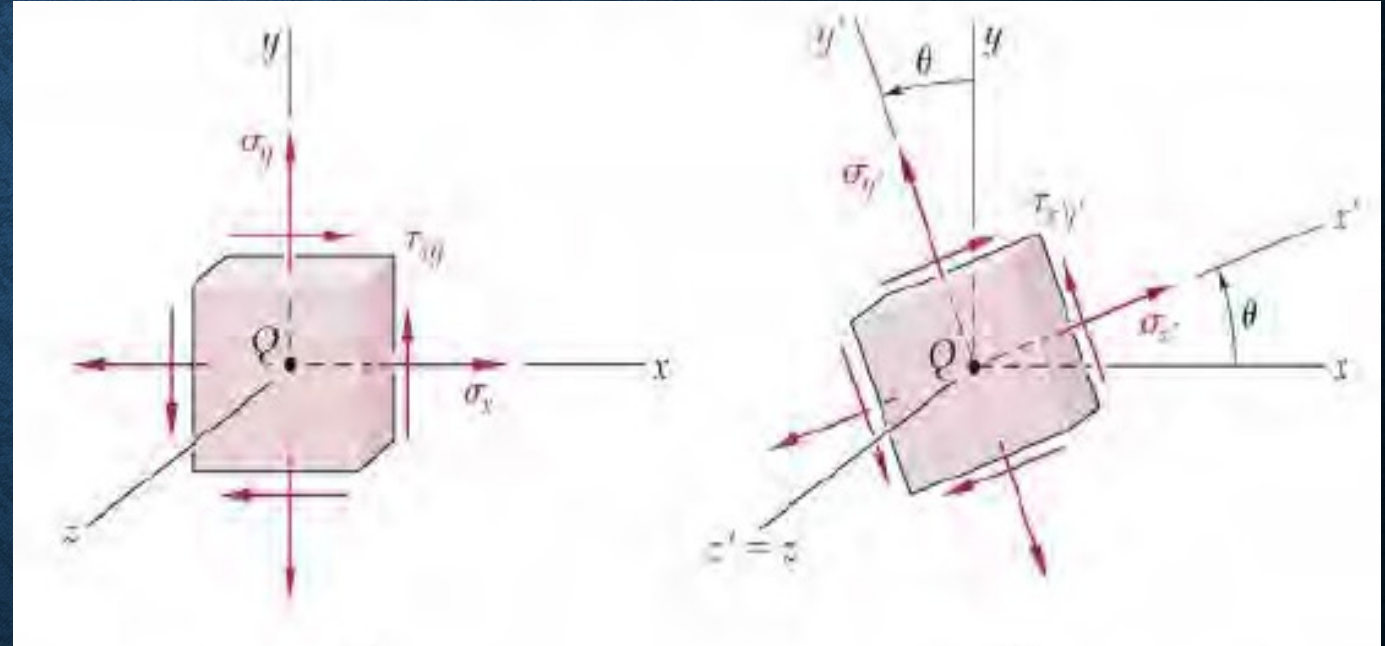
$$\sigma_z = \tau_{zx} = \tau_{zy} = 0$$

- State of plane stress occurs :
 - in a thin plate subjected to forces acting in the mid plane of the plate and
 - on the free surface of a structural element or machine component, i.e., at any point of the surface not subjected to an external force



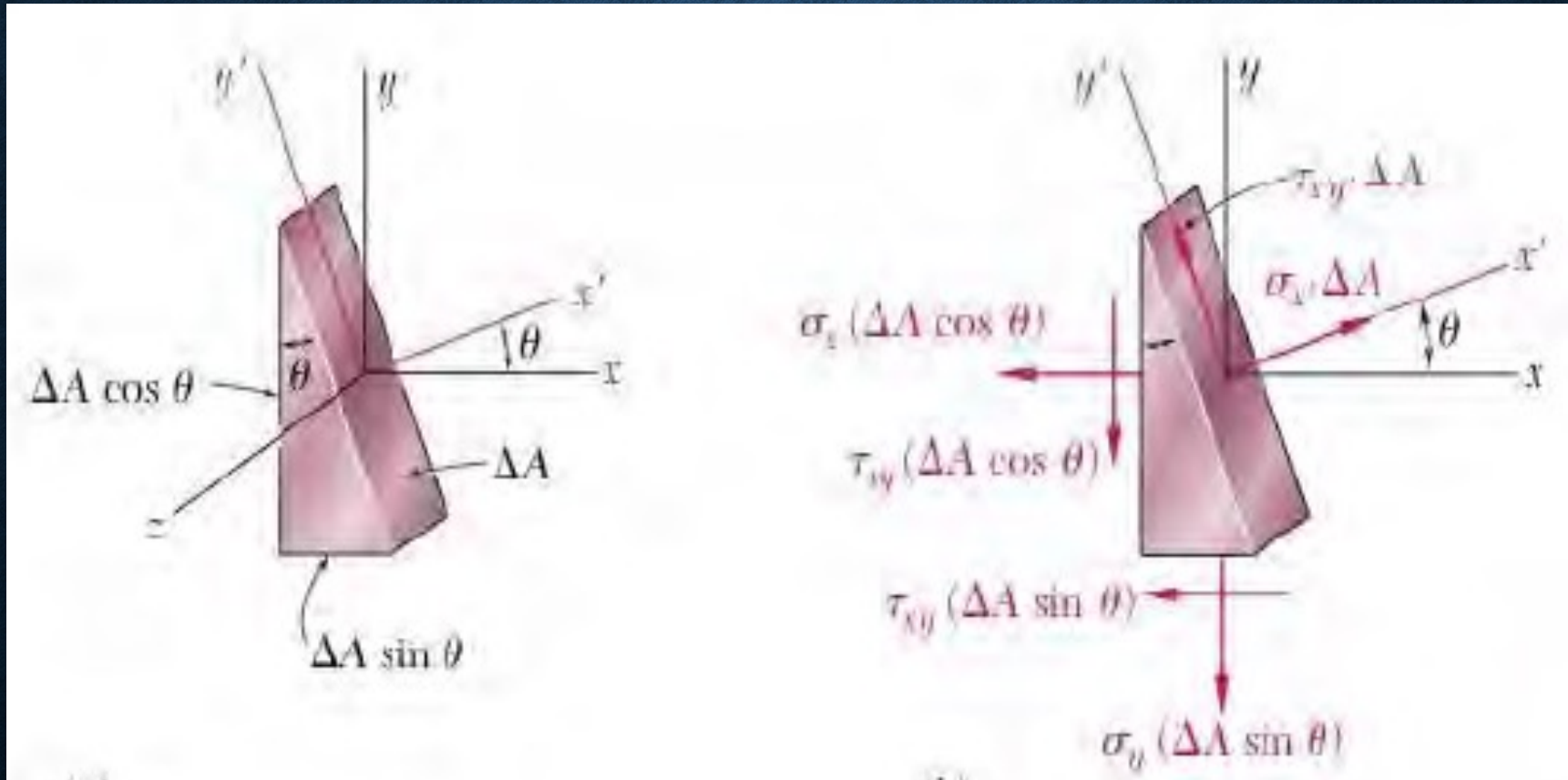
1.2 Stresses on Oblique Plane (Stress Transformation)

- Consider a state of plane stress exists at point Q (with $\sigma_z = \tau_{zx} = \tau_{zy} = 0$) and it is characterized by the stress components σ_x , σ_y , and τ_{xy} associated with the element shown below (left figure).



- The stress components $\sigma_{x'}$, $\sigma_{y'}$, and $\tau_{x'y'}$ associated with the element, after it has been rotated through an angle θ about the z axis (right figure), can be expressed in terms of σ_x , σ_y , τ_{xy} and θ using stress transformation equation.

- To determine the normal stress $\sigma_{x'}$ and the shear stress $\tau_{x'y'}$ exerted on the face perpendicular to the x' axis, let's consider a prismatic element with faces respectively perpendicular to x, y and x' axes.



➤ Using the equilibrium equation along the x' and y' axes

$$\begin{aligned}\sum F_{x'} = 0 = & \sigma_{x'} \Delta A - \sigma_x (\Delta A \cos \theta) \cos \theta - \tau_{xy} (\Delta A \cos \theta) \sin \theta \\ & - \sigma_y (\Delta A \sin \theta) \sin \theta - \tau_{xy} (\Delta A \sin \theta) \cos \theta\end{aligned}$$

$$\begin{aligned}\sum F_{y'} = 0 = & \tau_{x'y'} \Delta A + \sigma_x (\Delta A \cos \theta) \sin \theta - \tau_{xy} (\Delta A \cos \theta) \cos \theta \\ & - \sigma_y (\Delta A \sin \theta) \cos \theta + \tau_{xy} (\Delta A \sin \theta) \sin \theta\end{aligned}$$

➤ Solving the first equation for $\sigma_{x'}$, and the second for $\tau_{x'y'}$, we have

$$\sigma_{x'} = \sigma_x \cos^2 \theta + \sigma_y \sin^2 \theta + 2\tau_{xy} \sin \theta \cos \theta \dots\dots\dots \text{(iii)}$$

$$\tau_{x'y'} = -(\sigma_x - \sigma_y) \sin \theta \cos \theta + \tau_{xy} (\cos^2 \theta - \sin^2 \theta) \dots\dots\dots \text{(iv)}$$

➤ Recalling the trigonometric relations

$$\sin 2\theta = 2 \sin \theta \cos \theta \qquad \cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

and

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2} \qquad \sin^2 \theta = \frac{1 - \cos 2\theta}{2}$$

and rearranging we write eqns. (iii) and (iv) as follows

$$\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta \dots\dots\dots (1.1)$$

$$\tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta \dots\dots\dots (1.2)$$

❖ These equations are usually called the **transformation equations for plane stress** because they transform the stress components from one set of axes (x, y) to another (x', y')

- The expression for the normal stress $\sigma_{y'}$, is obtained by replacing θ in Eq. (1.1) by the angle $\theta + 90^\circ$ that the y' axis forms with the x axis.

Since $\cos (2\theta + 180^\circ) = -\cos 2\theta$ and $\sin (2\theta + 180^\circ) = -\sin 2\theta$, we have

$$\sigma_{y'} = \frac{\sigma_x + \sigma_y}{2} - \frac{\sigma_x - \sigma_y}{2} \cos 2\theta - \tau_{xy} \sin 2\theta \dots\dots\dots (1.3)$$

- Adding Eqs. (1.1) and (1.3) member to member, we obtain

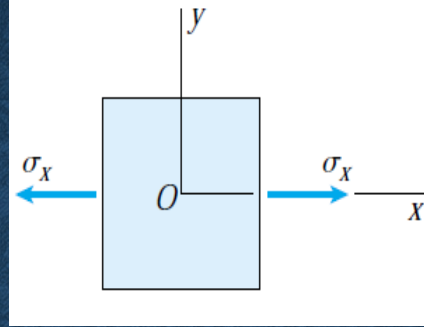
$$\sigma_{x'} + \sigma_{y'} = \sigma_x + \sigma_y \dots\dots\dots (1.4)$$

- ❖ Thus, in the case of plane stress the sum of the normal stresses exerted on a cubic element of material is independent of the orientation of that element.

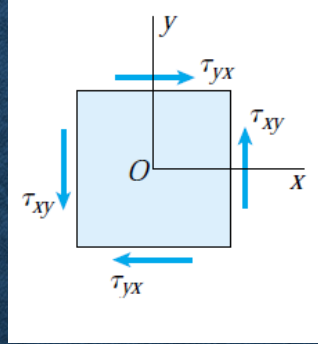
Special Case of Plane Stress

I) Uni-Axial State

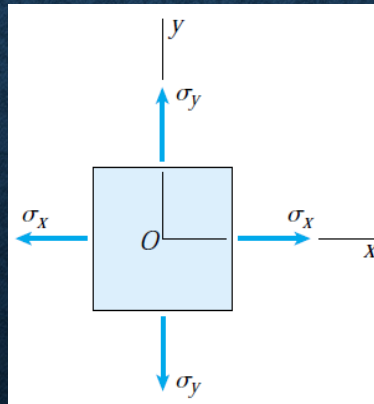
$$\sigma_{x'} = \sigma_x \cos^2 \theta$$



II) Pure Shear



III) Bi-Axial State (Pure Normal)



1.3 Principal Stresses and Maximum Shear Stress

- The transformation equations for plane stress (eqns. (1.1) and (1.2)) show that the normal stresses $\sigma_{x'}$ and the shear stresses $\tau_{x'y'}$ vary continuously as the axes are rotated through the angle θ .
- Both the normal and shear stresses reach maximum and minimum values within a given interval and these maximum and minimum values are usually needed for design purposes.

Principal Stresses

- The maximum and minimum normal stresses, called the **principal stresses**, can be found from the transformation equation for the normal stress $\sigma_{x'}$ (Eq. 1.1), by taking the derivative of $\sigma_{x'}$ with respect to θ and setting it equal to zero, we obtain an equation from which we can find the values of θ at which $\sigma_{x'}$ is a maximum or a minimum.

➤ Using Eqn. 1.1 and taking its first derivative with respect to θ

$$\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta$$

$$\frac{d\sigma_{x'}}{d\theta} = -(\sigma_x - \sigma_y) \sin 2\theta + 2\tau_{xy} \cos 2\theta = 0$$

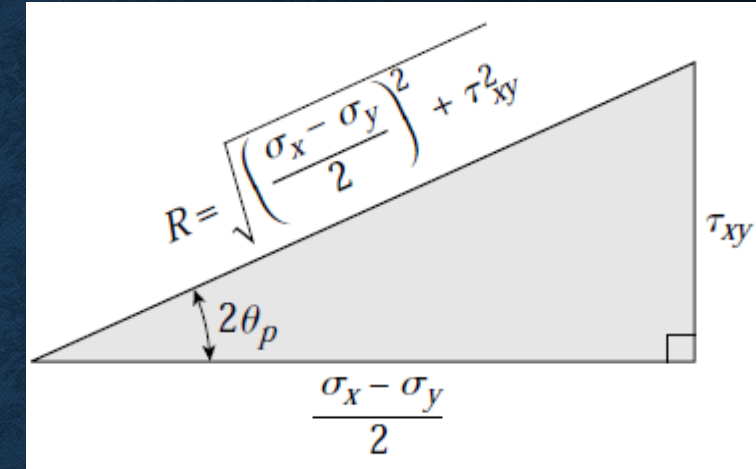
➤ From which we get the principal plane

$$\tan 2\theta_p = \frac{2\tau_{xy}}{\sigma_x - \sigma_y} \dots\dots\dots (1.5)$$

- The subscript p indicates that the angle θ_p defines the orientation of the **principal planes**, that is, the planes on which the principal stresses act.

- Let's refer to the right triangle in the figure below, which is constructed from Eq. (1.5). Note that the hypotenuse of the triangle, obtained from the Pythagorean theorem, is

$$R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$



- From the triangle we obtain two additional relations:

$$\cos 2\theta_p = \frac{\sigma_x - \sigma_y}{2R} \quad \sin 2\theta_p = \frac{\tau_{xy}}{R}$$

- Substituting these expressions for $\cos 2\theta$ and $\sin 2\theta$ into Eq. (1.1) and obtain the algebraically larger of the two principal stresses, denoted by σ_1 :

$$\begin{aligned} \sigma_1 = \sigma_{x_1} &= \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta_p + \tau_{xy} \sin 2\theta_p \\ &= \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \left(\frac{\sigma_x - \sigma_y}{2R} \right) + \tau_{xy} \left(\frac{\tau_{xy}}{R} \right) \end{aligned}$$

➤ After substituting for R and performing some algebraic manipulations, we obtain

$$\sigma_1 = \frac{\sigma_x + \sigma_y}{2} + \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \dots\dots\dots (v)$$

➤ The smaller of the principal stresses, denoted by σ_2 , may be found from the condition that the sum of the normal stresses on perpendicular planes is constant (see Eq. 1.4):

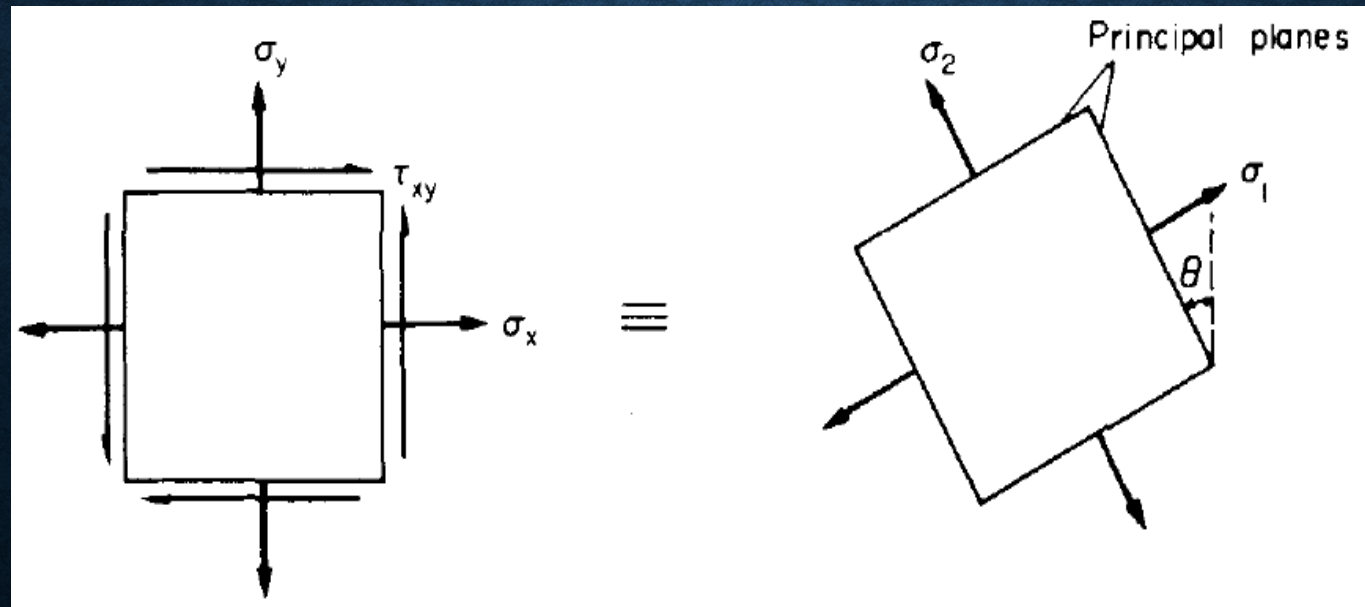
$$\sigma_1 + \sigma_2 = \sigma_x + \sigma_y \dots\dots\dots (1.4)$$

➤ Substituting the expression for σ_1 into Eq. (1.4) and solving for σ_2 , we get

$$\begin{aligned} \sigma_2 &= \sigma_x + \sigma_y - \sigma_1 \\ &= \frac{\sigma_x + \sigma_y}{2} - \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \dots\dots\dots (vi) \end{aligned}$$

➤ Combining Eqns. (v) and (vi) in to a single formula the famous *Principal Stresses* formula can be expressed as

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$



Maximum Shear Stress

- The maximum Shear Stress can be found from the transformation equation for the shear stress $\tau_{x'y'}$ (Eq. 1.2), by taking the derivative of $\tau_{x'y'}$ with respect to θ and setting it equal to zero, we obtain an equation from which we can find the values of θ at which $\tau_{x'y'}$ is a maximum.
- Using Eqn. 1.2 and taking its first derivative with respect to θ

$$\tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta$$

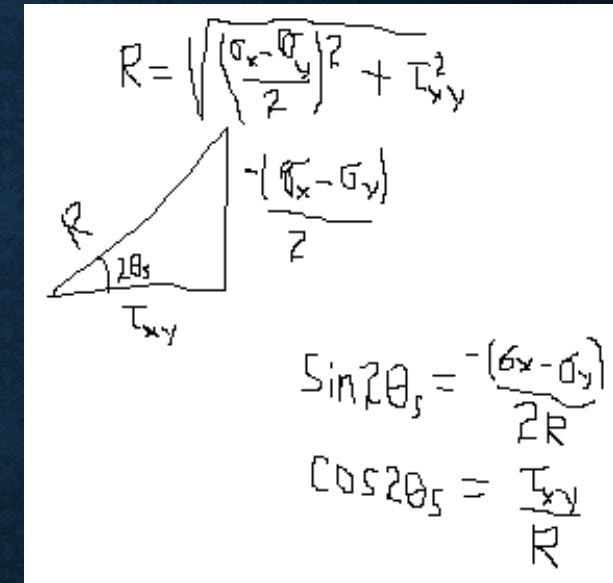
$$\frac{d\tau_{x'y'}}{d\theta} = -(\sigma_x - \sigma_y) \cos 2\theta - 2\tau_{xy} \sin 2\theta = 0$$

- From which we get the plane at which the shear stress is maximum

$$\tan 2\theta_s = -\frac{\sigma_x - \sigma_y}{2\tau_{xy}} \dots\dots\dots (1.7)$$

- The subscript s indicates that the angle θ_s defines the orientation of the maximum shear **plane**, that is, the planes on which the maximum shear stress acts.
- From Eqn. (1.7) we can construct a right triangle as shown in the figure below.
- Substituting these expressions for $\cos 2\theta$ and $\sin 2\theta$ into Eq. (1.2), the maximum shear stress can be obtained.

$$\tau_{\max} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \dots\dots\dots (1.8)$$



- Equating Eqn. (1.5) with Eqn. (1.7) we will get a relationship between principal plane θ_p and maximum shear plane θ_s .

$$\theta_s = \theta_p - 45^\circ \dots\dots\dots(1.9)$$

1.4 Mohr's Circle of Stress

- The transformation equations for plane stress can be represented in graphical form by a plot known as **Mohr's circle**.
- **Mohr's Circle of Stress:**
 - Enables to visualize the relationships between the normal and shear stresses acting on various inclined planes at a point in a stressed body
 - Provides a means for calculating principal stresses, maximum shear stresses, and stresses on inclined planes

Equations of Mohr's Circle

- Can be derived from the two transformation equations for plane stress, (Eqns. (1) and (2)), with a slight rearrangement of the first equation.

$$\sigma_{x'} - \frac{\sigma_x + \sigma_y}{2} = \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta$$

$$\tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta \dots\dots\dots \text{(vii)}$$

- Squaring both sides of each equation and then adding the two equations to eliminate the parameter 2θ ,

$$\left(\sigma_{x'} - \frac{\sigma_x + \sigma_y}{2}\right)^2 + \tau_{x'y'}^2 = \left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2 \dots\dots\dots (viii)$$

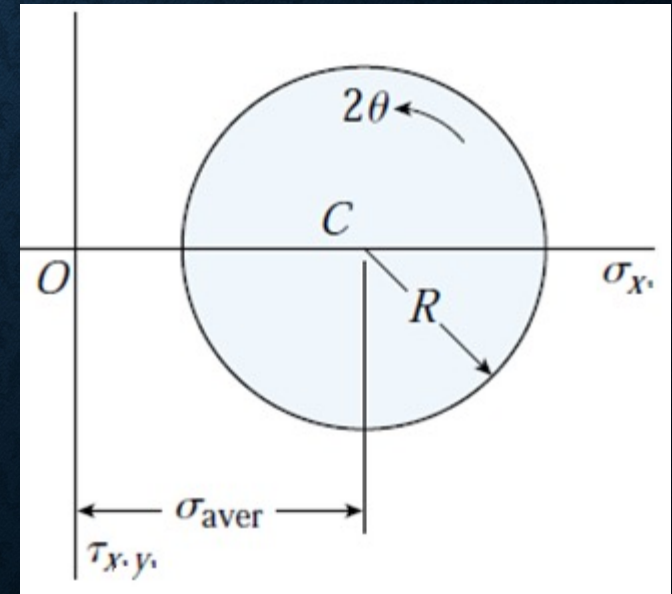
- Representing *some terms* by σ_{aver} and R ,

$$\sigma_{aver} = \frac{\sigma_x + \sigma_y}{2} \quad R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \dots\dots\dots (ix)$$

the equation of the Mohr's Circle can be written as

$$(\sigma_{x'} - \sigma_{aver})^2 + \tau_{x'y'}^2 = R^2 \dots\dots\dots (1.10)$$

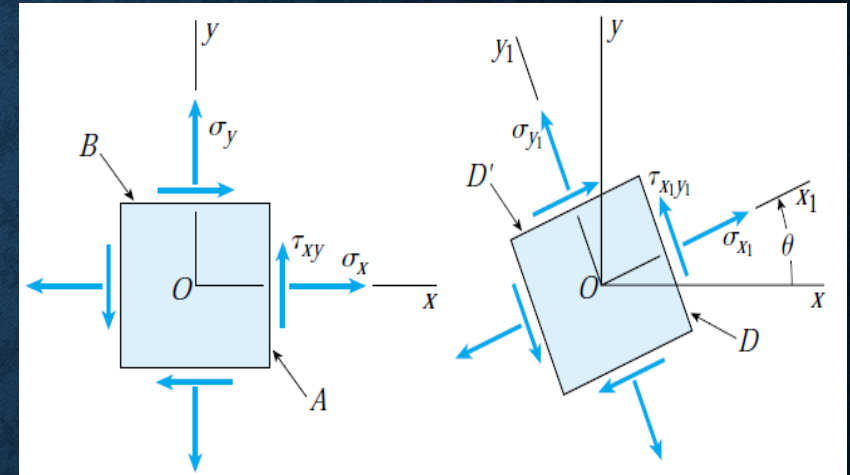
Which is the equation of a circle,
with horizontal axis σ' and vertical axis $\tau_{x'y'}$,
and its center $(0, \sigma_{aver})$ and radius R



Construction of Mohr's Circle

➤ With σ_x , σ_y and τ_{xy} known, the procedure for constructing Mohr's circle is as follows

1. Draw a set of coordinate axes with σ_x as abscissa (positive to the right) and τ_{xy} as ordinate (positive downward)
2. Locate the center C of the circle at the point having coordinates $\sigma_x = \sigma_{ave}$ and $\tau_{xy} = 0$
3. Locate point A , representing the stress conditions on the x face of the element shown in Fig. 7-15a, by plotting its coordinates $\sigma_x = \sigma_x$ and $\tau_{xy} = \tau_{xy}$.



Note that point A on the circle corresponds to $\theta = 0$.

Also, note that the x face of the element (Fig. 7-15a) is labeled “ A ” to show its correspondence with point A on the circle.

4. Locate point B , representing the stress conditions on the y face of the element shown in Fig. 7-15a, by plotting its coordinates $\sigma_{x'} = \sigma_y$ and $\tau_{x'y'} = -\tau_{xy}$. Note that point B on the circle corresponds to $\theta = 90^\circ$. In addition, the y face of the element (Fig. 7-15a) is labeled “ B ” to show its correspondence with point B on the circle.

6. Using point C as the center, draw Mohr's circle through points A and B . The circle drawn in this manner has radius R

